Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The value of $\lim_{n\to\infty} \frac{1}{n}\left(\sin\frac{\pi}{n}+\sin\frac{2\pi}{n}+\cdots+\sin\frac{n\pi}{n}\right)$ is

(A) π

(B) $-\frac{\pi}{2}$

(C) $\frac{1}{\pi}$

(D) $\frac{2}{\pi}$

Q.2

If $f(n)=\frac{1}{n}[(n+1)(n+2)\ldots(n+n)]^{1/n}$, then $\lim_{n\to\infty} f(n)$ equals

(A) e

(B) $1/e$

(C) $2/e$

(D) 4/e

Q.3

The value of $\lim_{n\to\infty}\left(\frac{1}{n}+\frac{n}{(n+1)^2}+\frac{n}{(n+2)^2}+\cdots+\frac{n}{(2n-1)^2}\right)$ is

(A) 1

(B) $1/3$

(C) $1/2$

(D) $3/2$

Q.4

The value of $\lim_{n\to\infty}\left(\frac{1}{1^3+n^3}+\frac{2^2}{2^3+n^3}+\cdots+\frac{n^2}{n^3+n^3}\right)$ is

(A) $\frac{1}{3}$

(B) $\frac{1}{3}\log 2$

(C) $\frac{1}{2}\log 3$

(D) $\frac{1}{3}\log 3$

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Q.5

$\lim_{n\to\infty}\left(\frac{(n+1)(n+2)\cdots(3n)}{n^{2n}}\right)^{1/n}$ is equal to:

(A) $\frac{18}{e^4}$

(B) $\frac{27}{e^2}$

(C) $\frac{9}{e^2}$

(D) $3\log 3-2$

Q.6

The value of $\lim_{m\to\infty}\frac{\int_0^{\pi/2}\sin^{2m}x\,\mathrm{d}x}{\int_0^{\pi/2}\sin^{2m+1}x\,\mathrm{d}x}$

(A) 0

(B) $1/2$

(C) 2

(D) 1

Q.7

The value of $\int_0^{\pi}\sin^n x\cos^{2m+1}x\,\mathrm{d}x$ is

(A) $\frac{(2m+1)!}{(n!)^2}$

(B) $\frac{(2m+1)!}{n!}$

(C) $\int_0^{\pi}\cos^{2m-1}x\,\mathrm{d}x$

(D) none of these

Q.8

If $a_n=\int_0^{\frac{\pi}{2}}\frac{\sin^2 nx}{\sin x}dx$ then
$a_2-a_1, a_3-a_2,\quad a_4-a_3, \ldots\ldots\ldots\ldots$ are in

(A) A.P.

(B) GP.

(C) H.P.

(D) None of these

Q.9

The value of the integral $\int_a^b\frac{|x|}{x}dx$, where $a<b$ is

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(A) $b - a$

(B) $|b| - |a|$

(C) $a + b$

(D) $|a| + |b|$

Q.10

$\left[\sum_{n=1}^{10} \int_{-2n-1}^{-2n} \sin^{27} x dx\right] + \left[\sum_{n=1}^{10} \int_{2n}^{2n+1} \sin^{27} x dx\right] =$

(A) 27^2

(B) -54

(C) 54

(D) 0

Q.11

Let $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$ and $K = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$. Then which one of the following is true ?

(A) $K < 2/3$ and $J < 2$

(B) $K > 2/3$ and $J > 2$

(C) $K < 2/3$ and $J > 2$

(D) $K > 2/3$ and $J < 2$

Q.12

$\int_0^\infty \left[\frac{2}{e^x}\right] dx$ is equal to ($[x]$ = greatest integer $\leq x$)

(A) $\log_e 2$

(B) e^2

(C) $\underline{2}$

(D) 0

Q.13

If $I_1 = \int_0^{\frac{\pi}{2}} \cos(\sin x) dx; I_2 = \int_0^{\frac{\pi}{2}} \sin(\cos x) dx$ and $I_3 = \int_0^{\frac{\pi}{2}} \cos x dx$, then

(A) $I_1 > I_3 > I_2$

(B) $I_3 > I_1 > I_2$

(C) $I_1 > I_2 > I_3$

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(D) $I_3 > I_2 > I_1$

Q.14

$\lim_{n\to\infty}\left[\frac{1}{2n+1}+\frac{1}{2n+2}+\ldots.+\frac{1}{2n+n}\right]=$

(A) $\log_e \frac{1}{3}$

(B) $\log_e \frac{2}{3}$

(C) $\log_e \frac{3}{2}$

(D) $\log_e \frac{4}{3}$

Q.15

$\text{Lim}_{n\to\infty}\left\{\frac{n!}{(kn)^n}\right\}^{\frac{1}{n}}$, where $k \neq 0$ is a constant and $n \in \mathbf{N}$ is equal to

(A) ke

(B) $k^{-1}e$

(C) ke^{-1}

(D) $k^{-1}e^{-1}$

Answers & Solutions

Q.1 Answer:

$\frac{2}{\pi}$

Solution:

$$\text{Reqd. limit } = \lim_{n\to\infty}\frac{1}{n}\sum_{n=1}^{n}\sin\frac{r\pi}{n}$$
$$= \int_0^1 \sin \pi x \, dx = -\left.\frac{\cos \pi x}{\pi}\right|_0^1$$
$$= \frac{-1}{\pi}[-1-1] = \frac{2}{\pi}$$

Q.2 Answer:

4/e

Solution:

$$A = \lim_{n\to 0} f(n)$$
$$= \lim_{n\to\infty}\left[\left(\frac{n+1}{n}\right)\left(\frac{n+2}{n}\right)\cdots\left(\frac{n+n}{n}\right)\right]^{\frac{1}{n}}$$
$$= \lim_{n\to\infty}\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\cdots\left(1+\frac{n}{n}\right)\right]^{\frac{1}{n}}$$
$$\log A = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\log\left(1+\frac{r}{n}\right) = \int_0^1 \log(1+x)dx$$
$$= \log\frac{4}{e}$$
$$\Rightarrow \quad A = \frac{4}{e}$$

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Q.3 Answer:

$1/2$

Solution:

$$\text{Required limit} = \lim_{n\to\infty} \sum_{r=0}^{n-1} \frac{n}{(n+r)^2}$$
$$= \lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{n-1} \frac{1}{\left(1+\frac{r}{n}\right)^2}$$
$$= \int_0^1 \frac{dx}{(1+x)^2} = -\frac{1}{1+x}\Big|_0^1.$$
$$= -\frac{1}{2} + 1 = \frac{1}{2}$$

Q.4 Answer:

$\frac{1}{3}\log 2$

Solution:

$$\text{Required limit} = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{r^2}{n^3+r^3}$$
$$= \lim_{n\to\infty} \frac{1}{n} \sum_{n=1}^{n} \frac{\left(\frac{r}{n}\right)^2}{1+\left(\frac{r}{n}\right)^3}$$
$$= \int_0^1 \frac{x^2}{1+x^3} dx = \frac{1}{3}\log 2.$$

Q.5 Answer:

$\frac{27}{e^2}$

Solution:

$$\text{Let } y = \lim_{x\to\infty} \left(\frac{(n+1)(n+2)\ldots(n+2n)}{n^{2n}}\right)^{1/n}$$
$$\log y = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{2n} \log\left(1+\frac{k}{h}\right)$$
$$= \int_0^2 \log(1+x)dx$$
$$= x\log(1+x)]_0^2 - \int_0^2 \frac{x}{1+x} dx$$
$$= 2\log 3 - \int_0^2 \frac{x+1-1}{x+1} dx$$
$$= 2\log 3 - 2 + \log(x+1)|_0^2$$
$$= 2\log 3 - 2 + \log 3$$
$$= \log 27 - \log e^2 = \log\frac{27}{e^2} \Rightarrow y = \frac{27}{e^2}.$$

Q.6 Answer:

1

Solution:

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We know that $I_{2n} = \int_0^{\pi/2} \sin^{2n} x \, dx$

$= \dfrac{2n-1}{2n} \times \dfrac{2n-3}{2n-2} \times \ldots \times \dfrac{1}{2} \times \dfrac{\pi}{2},$

$I_{2n+1} = \displaystyle\int_0^{\pi/2} \sin^{2n+1} x \, dx$

$= \dfrac{2n}{2n+1} \times \dfrac{2n-2}{2n-1} \times \ldots \times \dfrac{2}{3}$ and

Also, $I_{2m+1} = \frac{2m}{2m+1} I_{2m-1}$.

For all $x \in (0, \pi/2), \sin^{2m-1} x > \sin^{2m} x > \sin^{2m+1} x$

Integrating from 0 to $\pi/2$**, we get** $I_{2m-1} \geq I_{2m} \geq I_{2m+1}$

hence

$\frac{I_{2m-1}}{I_{2m+1}} \geq \frac{I_{2m}}{I_{2m+1}} \geq 1$ **(i)**

Also $\frac{I_{2m-1}}{I_{2m+1}} = \frac{2m+1}{2m}$.

Hence $\lim_{m\to\infty} \frac{I_{2m-1}}{I_{2m+1}} = \lim_{m\to\infty} \frac{2m+1}{2m} = 1.$

From (i) and using sandwitch theorem we have

$\lim_{m\to\infty} \frac{I_{2m}}{I_{2m+1}} = 1.$

Q.7 Answer:

$\int_0^{\pi} \cos^{2m-1} x \, dx$

Solution:

$I = \int_0^{\pi} \sin^n x \cos^{2m+1} x \, dx$

$= \int_0^{\pi} \sin^n(\pi - x) \cos^{2m+1}(\pi - x) dx$

$= -\int_0^{\pi} \sin^n x \cos^{2m+1} x \, dx = -I$

So $2I = 0$ **which implies that** $I = 0$ **Also,** $\int_0^{\pi} \cos^{2m-1} x \, dx = 0$ **by a similar reasoning.**

Q.8 Answer:

H.P.

Solution:

$a_n - a_{n-1} = \int_0^{\frac{\pi}{2}} \frac{\sin^2 nx - \sin^2(n-1)x}{\sin x} dx$

$= \int_0^{\frac{\pi}{2}} \frac{\sin\{nx+(n-1)x\} \sin\{nx-(n-1)x\}}{\sin x} dx$

[Using identity $\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$**]**

$= \int_0^{\frac{\pi}{2}} \frac{\sin(2nx-x)\sin x}{\sin x} dx = \int_0^{\frac{\pi}{2}} \sin(2n-1)x dx$

$= -\left[\frac{\cos(2n-1)x}{2n-1}\right]_0^{\frac{\pi}{2}} = -\left[\frac{\cos(2n-1)\frac{\pi}{2}}{2n-1} - \frac{\cos 0}{2n-1}\right]$

$= \frac{1}{2n-1}$

$\therefore a_2 - a_1 = \frac{1}{3}, a_3 - a_2 = \frac{1}{5}, a_4 - a_3 = \frac{1}{7} \ldots\ldots\ldots\ldots$ **etc.**

$\therefore a_2 - a_1, a_3 - a_2, \quad a_4 - a_3 \ldots\ldots\ldots\ldots$ **form an H.P.**

Q.9 Answer:

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$|b| - |a|$

Solution:

Three cases may arise

Case (i) : let $0 \leq a < b$, **then** $\frac{|x|}{x} = \frac{x}{x} = 1$

$\therefore \int_a^b \frac{|x|}{x} dx = \int_a^b dx = b - a = |b| - |a|$

(Since $|a| = a$ **and** $|b| = b$ **)**

Case (ii) : Let $a < b \leq 0$, **then** $\frac{|x|}{x} = \frac{-x}{x} = -1$

$\int_a^b \frac{|x|}{x} dx = \int_a^b (-1) dx = a - b = -|a| + |b|$

(Since $|a| = -a$ **and** $|b| = -b$ **)**

Case (iii) : Let $a < 0 < b$, **then** $\int_a^b \frac{|x|}{x} dx = \int_a^0 \frac{|x|}{x} dx + \int_0^b \frac{|x|}{x} dx$

$= \int_a^0 (-1) dx = \int_0^b (1) dx = a + b = -|a| + |b|$

(**Since** $|a| = -a$ **and** $|b| = b$) **Thus, in every case** $\int_a^b \frac{|x|}{x} dx = |b| - |a|$

Q.10 Answer:

0

Solution:

General term of the series $\sum_{n=1}^{10} \int_{-2n-1}^{-2n} \sin^{27} x dx$ **is**

$I_1 = \int_{-2n-1}^{-2n} \sin^{27} x dx = \int_{2n+1}^{2n} \sin^{27}(-x)(-dx)$

$= -\int_{2n}^{2n+1} \sin^{27} x dx = -I_2$

Where I_2 **is general term of series**

$\sum_{n=1}^{10} \int_{2n}^{2n+1} \sin^{27} x dx$

Q.11 Answer:

$K < 2/3$ and $J < 2$

Solution:

We have, $K = \int_0^1 \frac{\sin x}{\sqrt{x}} dx < \int_0^1 \frac{x}{\sqrt{x}} dx = \int_0^1 \sqrt{x} dx$

$= \frac{2}{3} \left[x^{3/2}\right]_0^1 = \frac{2}{3} \Rightarrow K < \frac{2}{3}$

Now, $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx < \int_0^1 \frac{1}{\sqrt{x}} dx = [2\sqrt{x}]_0^1 = 2$

$\Rightarrow J < 2$

Q.12 Answer:

$\log_e 2$

Solution:

We have, if $e^x > 2$, $\frac{2}{e^x} < 1$. **Also** $\frac{2}{e^x} > 0$

$\Rightarrow 0 < \frac{2}{e^x} < 1 \quad \therefore$ **If** $x > \log_e 2$, $\left[\frac{2}{e^x}\right] = 0$

Again if $0 < x < \log_e 2$ **then** $1 < e^x < 2$

$\Rightarrow 1 > \frac{1}{e^x} > \frac{1}{2} \Rightarrow 2 > \frac{2}{e^x} > 1$ **or** $1 < \frac{2}{e^x} < 2$

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$\therefore \left[\frac{2}{e^x}\right] = 1$

$$\therefore I = \int_0^{\infty} \left[\frac{2}{e^x}\right] dx = \int_0^{\infty} \left[2e^{-x}\right] dx$$

$$= \int_0^{\log 2} \left[2e^{-x}\right] dx + \int_{\log 2}^{\infty} \left[2e^{-x}\right] dx$$

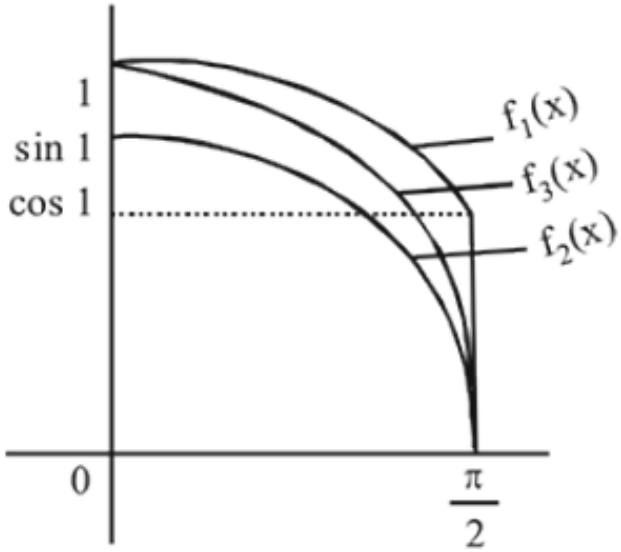
$$= \int_0^{\log 2} (1)dx + \int_{\log 2}^{\infty} (0)dx = \log_e 2$$

Q.13 Answer:

$I_1 > I_3 > I_2$

Solution:

$I_1 = \int_0^{\frac{\pi}{2}} \cos(\sin x)dx$



$I_3 = \int_0^{\frac{\pi}{2}} \cos x dx$

Let $f_1(x) = \cos(\sin x), f_2(x) = \sin(\cos x)$

$f_3(x) = \cos x$

If $x > 0$, **then** $\sin x < x$

$\Rightarrow$ **for** $0 < x < \frac{\pi}{2}, \sin(\cos x) < \cos x$

Also, $0 < x < \frac{\pi}{2}$ **then** $\sin x < x$

$\Rightarrow \cos(\sin x) > \cos x$

$\therefore \quad \cos(\sin x) > \cos x > \sin(\cos x)$ **if** $0 < x < \frac{\pi}{2}$

$\therefore \quad I_1 > I_3 > I_2$

Q.14 Answer:

$\log_e \frac{3}{2}$

Solution:

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$$\lim_{n\to\infty}\left[\frac{1}{2n+1}+\frac{1}{2n+2}+\ldots+\frac{1}{2n+n}\right]$$
$$=\lim_{n\to\infty}\frac{1}{n}\left[\frac{1}{2+\frac{1}{n}}+\frac{1}{2+\frac{2}{n}}+\ldots+\frac{1}{2+\frac{n}{n}}\right]$$
$$=\lim_{n\to\infty}\frac{1}{n}\sum_{n=1}^{n}\frac{1}{2+\frac{2}{n}}=\int_0^1\frac{dx}{2+x}$$
$$=[\log(2+x)]_0^1]=\log 3-\log 2=\log\frac{3}{2}$$

Q.15 Answer:

$\mathrm{k}^{-1}\mathrm{e}^{-1}$

Solution:

$$\text{Let } \mathrm{P}=\mathrm{Lim}_{\mathrm{n}\to\infty}\left\{\frac{\mathrm{n}!}{(\mathrm{kn})^{\mathrm{n}}}\right\}^{\frac{1}{\mathrm{n}}}$$

Taking log of both the sides at the base e

$$\begin{aligned}\log_{\mathrm{e}}\ \mathrm{P}&=\mathrm{Lim}_{\mathrm{n}\to\infty}\frac{1}{\mathrm{n}}\log_{\mathrm{e}}\left\{\frac{\mathrm{n}!}{(\mathrm{kn})^{\mathrm{n}}}\right\}\\&=\mathrm{Lim}_{\mathrm{n}\to\infty}\frac{1}{\mathrm{n}}\log_{\mathrm{e}}\left\{\frac{1}{\mathrm{kn}}\cdot\frac{2}{\mathrm{kn}}\cdot\frac{3}{\mathrm{kn}}\ldots\ldots\ldots\ldots\ldots\ldots\ldots\cdot\frac{\mathrm{n}}{\mathrm{kn}}\right\}\\=\mathrm{Lim}_{\mathrm{n}\to\infty}&\ \frac{1}{\mathrm{n}}\left[\log\left(\frac{1}{\mathrm{kn}}\right)+\log\left(\frac{2}{\mathrm{kn}}\right)+\ldots\ldots\ldots\ldots+\log\left(\frac{\mathrm{n}}{\mathrm{kn}}\right)\right]\\&=\mathrm{Lim}_{\mathrm{n}\to\infty}\frac{1}{\mathrm{n}}\sum_{\mathrm{r}=1}^{\mathrm{n}}\log\left(\frac{\mathrm{r}}{\mathrm{kn}}\right)\\&=\int_0^1\log\left(\frac{\mathrm{x}}{\mathrm{k}}\right)\mathrm{dx}=\int_0^1(\log\mathrm{x}-\log\mathrm{k})\mathrm{dx}\\&=\int_0^1\log\mathrm{xdx}-\int_0^1\log\mathrm{kdx}\\&=[\mathrm{x}\log\mathrm{x}-\mathrm{x}]_0^1-\log\mathrm{k}[\mathrm{x}]_0^1\\&=[0-1-0+0]-\log\mathrm{k}=-1-\log\mathrm{k}\\&=-(\log\mathrm{e}+\log\mathrm{k})=-\log(\mathrm{ek})=\log\frac{1}{\mathrm{ek}}\end{aligned}$$

[**Value of $x\log x$ at $x=0$ is** $\mathrm{Lim}_{x\to0^+}\ x\log x=0$]

$\therefore\ \mathrm{P}=\frac{1}{\mathrm{ek}}=\mathrm{k}^{-1}\mathrm{e}^{-1}$

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